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$$-5 + 12i = (x + yi)^2 = x^2 - y^2 + 2xyi$$

$$\begin{cases} x^2 - y^2 = -5 \\ 2xy = 12 \end{cases} \Rightarrow \begin{cases} x^2 - y^2 = -5 \\ xy = 6 \end{cases} \Rightarrow \begin{cases} x^2 - y^2 = -5 \\ x^2(-y)^2 = 36 \end{cases}$$

$$\lambda^2 + 5\lambda - 36 = 0$$

$$\Delta = 5^2 - 4 \cdot (-36) = 169 = 13^2$$

$$\lambda_1 = \frac{-5 + 13}{2} = 4$$

$$\lambda_2 = \frac{-5 - 13}{2} = -9$$

$$x = \pm 5 \ y = \pm 3$$

$$x + yi \in \{2 + 3i, -2 - 3i\}$$

References